



VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CT GROUP

CHEMISTRY

9729/03

Paper 3 Free Response

17 September 2018

Candidates answer on separate paper.

2 hours

Additional Materials: Cover Page
 Answer Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

You must start the answer to each question on a fresh piece of writing paper.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, fasten all your work securely together.

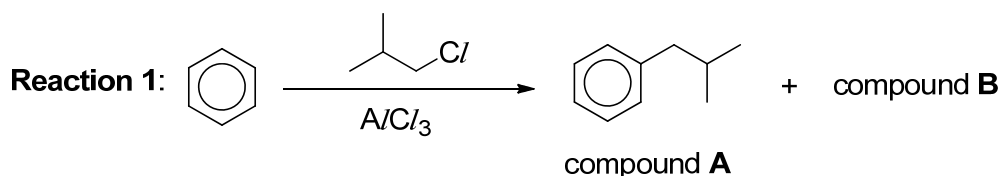
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

Section A

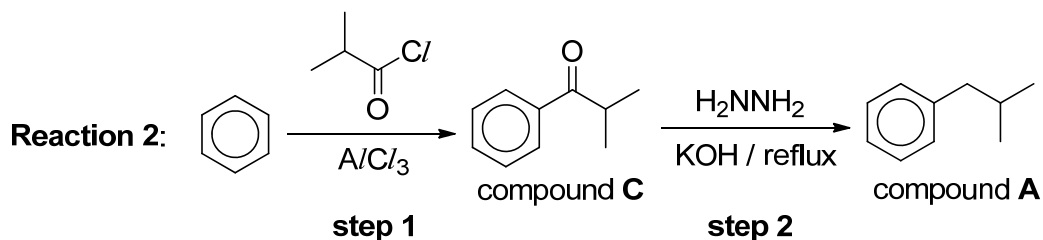
Answer **all** the questions in this section.

- 1 (a) All the elements in the third period of the Periodic Table, sodium to sulfur, form chlorides by direct combination with chlorine. Aluminium chloride may be produced by passing a stream of chlorine over heated aluminium metal in a long hard-glass tube.
- (i) With the aid of equations, explain the following observations when different amounts of water were added to solid aluminium chloride.
- (I) When a limited amount of water was added, a white solid was formed together with steamy fumes.
- (II) When excess water was added, a solution of pH 3 was obtained. [2]
- (ii) Both aluminium chloride and copper(I) complex solutions are colourless whereas a solution of copper(II) sulfate appears blue. Explain these observations. [3]
- (iii) Briefly describe the process of anodisation of aluminium. Write ion-electron equations for the reactions occurring at the anode and the cathode. [2]
- (b) Compound **A** can be synthesised from benzene using aluminium chloride via a simple Friedel–Crafts alkylation as shown in **Reaction 1**. In addition, compound **B**, an isomer of compound **A** is also formed.

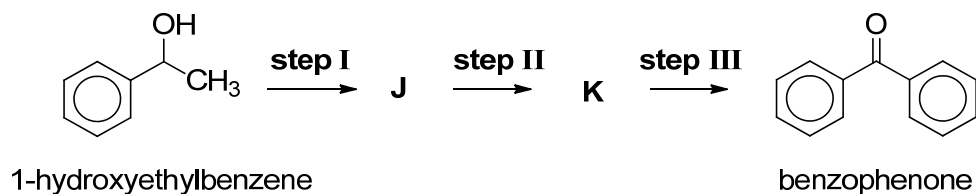


- (i) Compound **B** rotates plane-polarised light. It is formed after the carbocation intermediate undergoes rearrangement through the movement of an alkyl group to an adjacent carbon atom bearing the positive charge. Draw the structure of compound **B**. Explain why the rearrangement of the carbocation is favoured. [2]
- (ii) Explain why multi-substituted product is more favoured over mono-substituted product in **Reaction 1**. [1]

- (c) Compound **A** can be formed via compound **C** as shown in **Reaction 2** below. **Step 1** involves Friedel–Crafts acylation, which have similar reaction conditions and mechanism as Friedel-Crafts alkylation.



- (i) Draw the mechanism for **step 1**. In your answer, show relevant charges, lone pairs of electrons and movement of electrons. [3]
- (ii) Hence, suggest reagents and conditions for **steps I to III** in the following synthesis of benzophenone from 1-hydroxyethylbenzene. Give the structural formulae of **J** and **K**.



[4]

- (d) A student wants to synthesise benzophenone using the reaction pathway illustrated in (c)(ii). However, the solid sample of 1-hydroxyethylbenzene is contaminated with phenylamine. Briefly explain how you can separate 1-hydroxyethylbenzene from phenylamine via extraction. You are provided with

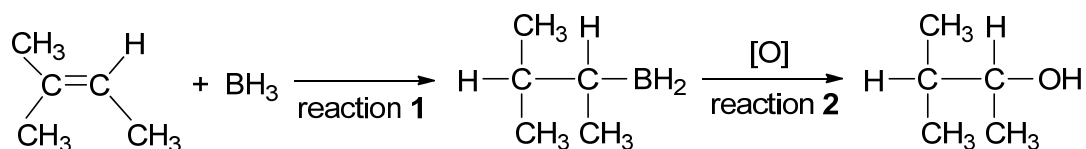
- ethanol, hexane, HCl(aq) , NaOH(aq) ,
- separating funnel, and
- apparatus commonly found in a college laboratory.

[3]

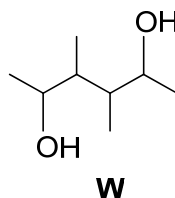
[Total: 20]

- 2 (a) Borane, BH_3 , is used to synthesise alcohols from alkenes as shown in the reaction sequence below.

In reaction 1, the BH_2 group from BH_3 is bonded to the **less** substituted carbon atom of the double bond. The remaining H atom from BH_3 is bonded to the other carbon atom.



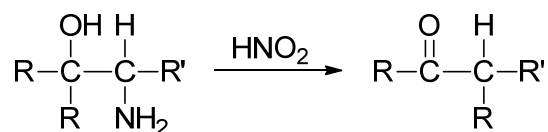
- (i) Name and show the mechanism of reaction 1. In your answer, show relevant charges, lone pairs of electrons and movement of electrons. [3]
- (ii) The product formed in reaction 1 contains a chiral centre. Explain whether the product formed rotates plane-polarised light. [2]
- (iii) The diol **W** can be prepared by the same method as shown above.



Draw the skeletal structure of the diene which could be used to prepare diol **W**. [1]

- (b) A carbonyl compound, **Y**, $\text{C}_6\text{H}_{10}\text{O}$, can be synthesised from aminoalcohol, **X**, $\text{C}_6\text{H}_{13}\text{ON}$, in the presence of nitrous acid, HNO_2 , via Tiffeneau-Demjanov Rearrangement.

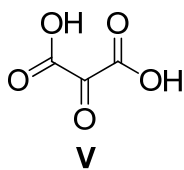
The simplified illustration of the rearrangement is shown below.



Y produces a yellow precipitate with iodine in alkaline solution. Treatment of **Y** with hot acidified solution of potassium manganate(VII) produces **Z**, $\text{C}_5\text{H}_8\text{O}_3$, along with a gas that forms a white precipitate in limewater. **Y** was also observed to decolourise bromine in tetrachloromethane readily.

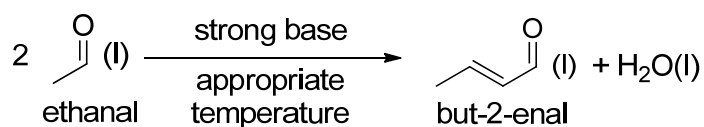
- (i) Explain the chemistry of the reactions described and deduce the structural formulae of **X**, **Y** and **Z**. [5]
- (ii) With reference to the *Data Booklet*, identify an infra-red absorption range that will be shown by the functional group in **Y** but **not** in **X**. [1]

- (c) Devise a three-step synthesis to obtain compound **V** from the carbonyl compound, CICH_2CHO .



Give the reagents and conditions for each step. Draw the structural formulae of all the intermediates. [4]

- (d) By either adjusting the reaction temperature or using a stronger base, ethanal can undergo Aldol Condensation reaction to form but-2-enal as shown below.

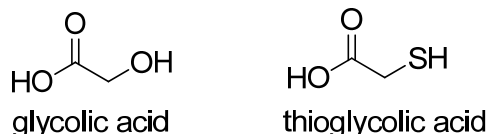


- (i) With reference to the *Data Booklet*, calculate the enthalpy change for this Aldol Condensation reaction. [2]
- (ii) Explain why the actual enthalpy change for the Aldol Condensation reaction is different from that calculated in (d)(i). [1]
- (iii) Suggest a simple chemical test to confirm the presence of but-2-enal in the product mixture. [1]

[Total: 20]

3 This question is about acids, bases and their derivatives.

(a) Glycolic acid and thioglycolic acid have very similar structures as shown below.



Glycolic acid has only one measurable pK_a value of 3.38. Thioglycolic acid has two measurable pK_a values of 3.67 and 10.31 respectively.

- (i) Both oxygen and sulfur are Group 16 elements. Hence, suggest why the alcohol functional group of glycolic acid is a weaker acid than the $-SH$ group of thioglycolic acid. [1]
- (ii) Compare the first pK_a values of glycolic acid and thioglycolic acid. Explain its significance. [2]
- (iii) Suggest the major species present in solutions of thioglycolic acid with the following pH values. [2]
 - pH 0
 - pH 7
 - pH 14
- (iv) Assuming thioglycolic acid to be H_2A , calculate the percentage of each of the ionic species of thioglycolic acid present at pH 9.0. [3]
- (v) Hence, calculate the volume of $0.100 \text{ mol dm}^{-3}$ aqueous sodium hydroxide needed to form the solution in (iv) when added to 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ thioglycolic acid. [2]

(b) The hydroxide ion is the strongest possible base in aqueous solution, but in organic solvents it is possible to have other stronger bases. In 2016, Australian researchers announced the formation of an organic gas-phase di-anion, DEB^{2-} . It is the strongest base known.

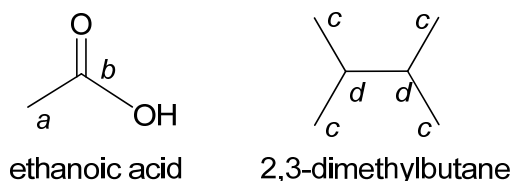
DEB^{2-} is produced from compound **A** which is a disubstituted benzene, $C_6H_4R_2$, where $R-$ is the same substituent.

- (i) Draw all the possible structures of $C_6H_4R_2$. [1]
- (ii) Compound **A** has the molecular formula $C_{12}H_6O_4$ and effervesces on addition of sodium hydrogen carbonate.

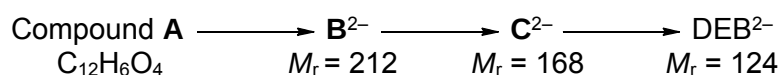
Suggest the functional group present in the substituent $R-$ that is responsible for the effervescence. Hence, deduce a structure for $R-$. [1]

^{13}C nuclear magnetic resonance (NMR) spectroscopy is an analytical method. In simplistic use, it allows different types of carbon atoms to be deduced from the corresponding signals in the spectrum.

For example, the ^{13}C NMR spectra for the ethanoic acid and 2,3-dimethylbutane would both show two distinct signals since each molecule has only two different carbon environments. This is shown in the diagram below where equivalent carbon environments are labelled with the same letter.



- (iii) Compound **A** is found to have six signals in its ^{13}C NMR spectrum. Using your answers to (b)(i), (b)(ii) and the information provided above, deduce the structure of compound **A**. [1]
- (iv) Compound **A** forms DEB^{2-} via intermediates B^{2-} and C^{2-} through the removal of positive ions or neutral molecule from the substituent R^- .



Determine the skeletal structures of B^{2-} , C^{2-} and DEB^{2-} . [2]

- (c) An organic acid **D**, $\text{C}_7\text{H}_{10}\text{O}_2$, is refluxed with acidified KMnO_4 to produce only one organic compound **E**, $\text{C}_4\text{H}_6\text{O}_4$. Compound **D** decolourises bromine water. Gentle heating of the anhydrous crystals of compound **E** produces a neutral compound **F**, $\text{C}_4\text{H}_4\text{O}_3$, which does not react with sodium metal or give a precipitate with 2,4-dinitrophenylhydrazine.

Compound **F** reacts with ethanol in the presence of a catalyst to form compound **G**, $\text{C}_6\text{H}_{10}\text{O}_4$.

Compound **F** also reacts with methylamine, CH_3NH_2 , to give a compound **H**, $\text{C}_5\text{H}_9\text{NO}_3$, which gives a salt on reaction with NaOH but not with HCl . Compound **H** does not display enantiomerism.

Both compounds, **G** and **H**, produce **E** on heating with dilute sulfuric acid.

Suggest the structures of **D**, **E**, **F**, **G** and **H**. [5]

[Total: 20]

Section B

Answer **one** question from this section.

- 4 (a) Chlorine dioxide, ClO_2 , is a reddish-yellow gas that does not occur naturally in the environment. When added to aqueous hydroxide, chlorine dioxide undergoes the following reaction producing chlorate(III) and chlorate(V) ions.

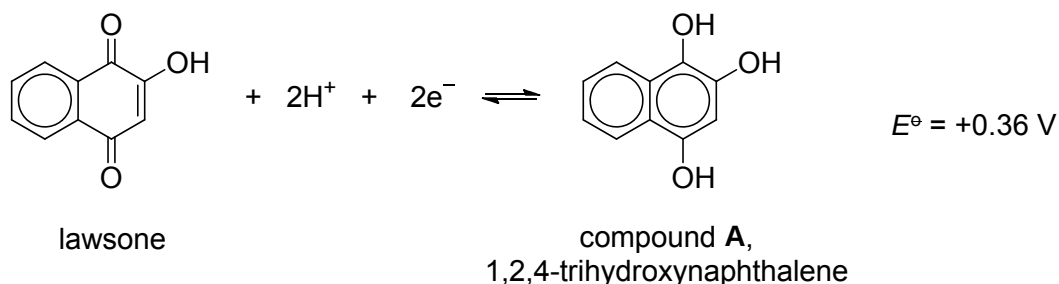


Unlike ClO_2 , bromine dioxide is less stable and react with aqueous hydroxide to produce bromide and bromate(V) ions. Bromate(V) ion is formed more readily than chlorate(V) ion.

- (i) Write a balanced equation for the reaction of bromine dioxide with sodium hydroxide. Hence, state the type of reaction. [2]
- (ii) Explain why bromate(V) ion is formed more readily than the chlorate(V) ion in aqueous hydroxide. [1]

Lawsone is a dye that is extracted from the henna plant, *Lawsonia inermis*. Although its natural colour is yellow, lawsone reacts with the proteins in hair and skin to produce the characteristic brown henna colour.

Lawsone can be readily reduced to a colourless compound **A**, 1,2,4-trihydroxynaphthalene. However, the hair dye containing lawsone is relatively resistant to the decolourisation by chlorine water in swimming pools.



- (iii) With reference to the *Data Booklet*, explain why chlorine water cannot spontaneously decolourise lawsone. [2]
- (iv) The Gibb's free energy change, ΔG° , is related to the equilibrium constant, K_c , through the following equation:

$$\Delta G^\circ = -RT \ln K_c$$

where ΔG° is in J mol^{-1} .

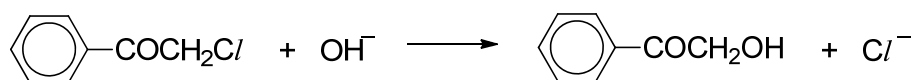
Using the answer from (a)(iii) and the *Data Booklet*, calculate K_c . Hence, deduce whether the position of equilibrium lies more on the left or right hand side of the equilibrium under standard conditions. [2]

- (v) Suggest a reagent that could be used to convert lawsone into compound **A** in the laboratory. [1]

- (vi) When lawsone is reacted with $\text{Br}_2(\text{aq})$, compound **B** with molecular formula, $\text{C}_{10}\text{H}_5\text{O}_3\text{Br}$, is formed. **B** reacts with 3 mol of 2,4-DNPH. Reaction of **B** with $\text{NaOH}(\text{aq})$ gives **C**. Compound **C** reacts with HCl and ZnCl_2 to produce **D**. The reaction of **D** with methylamine gives **E** with molecular formula, $\text{C}_{11}\text{H}_9\text{O}_3\text{N}$, which further reacts with CH_3COCl to give a neutral compound **F**.

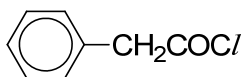
Suggest the skeletal formulae of compounds **B**, **C**, **D**, **E** and **F**. [5]

- (b) Chloroacetophenone was formerly the most widely used tear gas, under the codename CN. It was used in warfare and in riot control. Residues of CN can be destroyed by the hydrolysis of alkali.

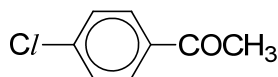


chloroacetophenone

G and **H** are isomers of chloroacetophenone.



G



H

- (i) Arrange the isomers, chloroacetophenone, **G** and **H**, in increasing ease of hydrolysis. Explain your choice. [3]
- (ii) Iodoacetophenone is even more reactive than chloroacetophenone towards alkaline hydrolysis. Briefly explain why it is so. [1]
- (iii) Suggest a suitable experimental technique for studying the rate of hydrolysis. [1]
- (iv) Compound **H** reacts with alkaline aqueous iodine to give a yellow solid, CHI_3 . Similar reaction occurs between compound **H** and alkaline aqueous chlorine.

Write a balanced equation for the reaction of compound **H** with alkaline aqueous chlorine. State the observation. [2]

[Total: 20]

- 5 (a) Copper can be extracted from a double salt, CuFeS_2 by reacting it with oxygen and silicon dioxide. In this reaction, copper, iron(II) silicate, FeSiO_3 , and an oxide of sulfur are produced. The oxide of sulfur does not decolourise acidified potassium manganate(VII).
- (i) Draw a dot-and-cross diagram for CuFeS_2 . [1]
- (ii) Write a balanced equation for the overall reaction. Deduce the reducing agent in the reaction. [2]
- (iii) Copper(I) sulfide is one of the reaction intermediate. Write the full electronic configuration of copper(I). [1]
- (iv) The crude copper obtained contains trace amount of silver and nickel. With relevant data from the *Data Booklet* and a labelled diagram, describe how the crude copper can be purified through electrolysis. [4]
- (v) 5.0 g of crude copper containing 0.1 g of silver and 0.1 g of nickel is purified through electrolysis as in (a)(iv). During electrolysis, 80% of the current is used to oxidise the metals. With reference to the *Data Booklet*, determine the time needed to purify the copper completely using a current of 2 A. [2]
- (vi) A student accidentally added some iron(III) nitrate into a solution of copper(II) nitrate. You are provided with
- $\text{NH}_3(\text{aq})$, $\text{NaOH}(\text{aq})$, $\text{HNO}_3(\text{aq})$ and
 - apparatus commonly found in a college laboratory.

Describe how the student could separate these two metal ions in the mixture. Give relevant equations and state the types of reactions that have occurred. [3]

- (b) Compounds **W**, **X**, **Y** and **Z** are constitutional isomers with molecular formula $C_9H_{10}O_2$ and none of them shows *cis-trans* isomerism. All four isomers can form intra-molecular hydrogen bonding. The results of seven tests carried out on the four isomers are shown below.

	Test	W	X	Y	Z
1	Rotate plane-polarised light	Yes	No	Yes	Yes
2	Add dilute NaOH at room temperature	Soluble	Soluble	Soluble	Insoluble
3	Heat with acidified $K_2Cr_2O_7$	Orange solution turned green	Orange solution did not turn green	Orange solution turned green	Orange solution turned green
4	Add 2,4-DNPH	Orange precipitate	Orange precipitate	No orange precipitate	Orange precipitate
5	Warm with Fehling's solution	Brick-red precipitate	No brick-red precipitate	No brick-red precipitate	No brick-red precipitate
6	Warm with Tollens' reagent	Grey ppt	No grey ppt	No grey ppt	Grey ppt
7	Warm with $I_2(aq)$ and NaOH(aq)	No pale yellow precipitate	Pale yellow precipitate	No pale yellow precipitate	Pale yellow precipitate

- (i) Use the information in the table above to suggest a skeletal formula for each of these four isomers. [4]
- (ii) Write a balanced equation for the reaction between **W** and 2,4-DNPH. [1]
- (iii) Explain why **W** is soluble in NaOH(aq) at room temperature. [2]

[Total: 20]

End of Paper